ECON 470 Homework 4

Submission 3, Spring 2026

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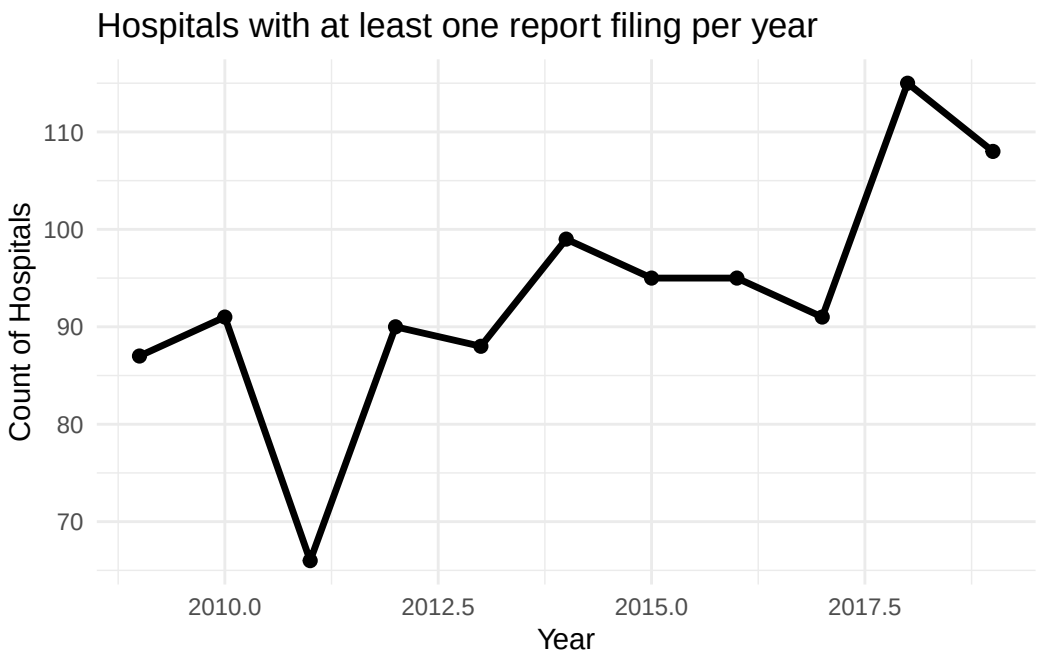
Data Filtering

```
# A tibble: 832 x 48
   provider_number fy_start   fy_end     date_processed date_created data_source
   <chr>           <date>     <date>     <date>         <date>       <chr>
 1 010012          2023-04-01 2024-12-31 2025-06-06     2025-06-03   <NA>
 2 010016          2009-10-01 2010-12-31 2011-06-28     2011-02-25   <NA>
 3 010024          2009-07-01 2010-12-31 2014-10-22     2011-01-25   <NA>
 4 010034          2002-07-01 2003-09-30 2005-03-15     2004-10-13   <NA>
 5 010038          2016-07-01 2017-09-30 2018-11-14     2018-03-20   <NA>
 6 010046          2014-07-01 2015-12-31 2016-07-25     2015-10-23   <NA>
 7 010054          2010-07-01 2011-12-31 2014-07-23     2013-11-12   <NA>
 8 010055          2013-07-01 2014-09-30 2015-03-20     2014-12-17   <NA>
 9 010059          2008-03-01 2009-09-30 2013-06-07     2010-04-07   <NA>
10 010064          2005-07-01 2006-12-31 2013-03-26     2012-10-23   <NA>
# i 822 more rows
# i 42 more variables: beds <dbl>, tot_charges <dbl>, net_pat_rev <dbl>,
#   tot_discounts <dbl>, tot_operating_exp <dbl>, ip_charges <dbl>,
#   icu_charges <dbl>, ancillary_charges <dbl>, tot_discharges <dbl>,
#   mcare_discharges <dbl>, mcaid_discharges <dbl>, tot_mcare_payment <dbl>,
#   secondary_mcare_payment <dbl>, street <chr>, city <chr>, state <chr>,
#   zip <chr>, county <chr>, name <chr>, uncomp_care <dbl>, ...
```

Upon dataset creation, prices were calculated using hospital cost report variables where negative prices were dropped, and top & bottom 1% outliers were removed. This was then used to curate data to answer questions for estimating ATEs. Price changes from 2011 through 2014 resulted from a simple difference between the two years; the net penalty was the same, with a difference between HVBP and HRRP payments. All these were reported alongside mean numbers for pre-period Medicare discharges, bed size, and Medicaid discharges from 2009 to 2011.

Problem 1

The number of hospitals filing more than one report is noticeable and varies over the years.



Problem 2

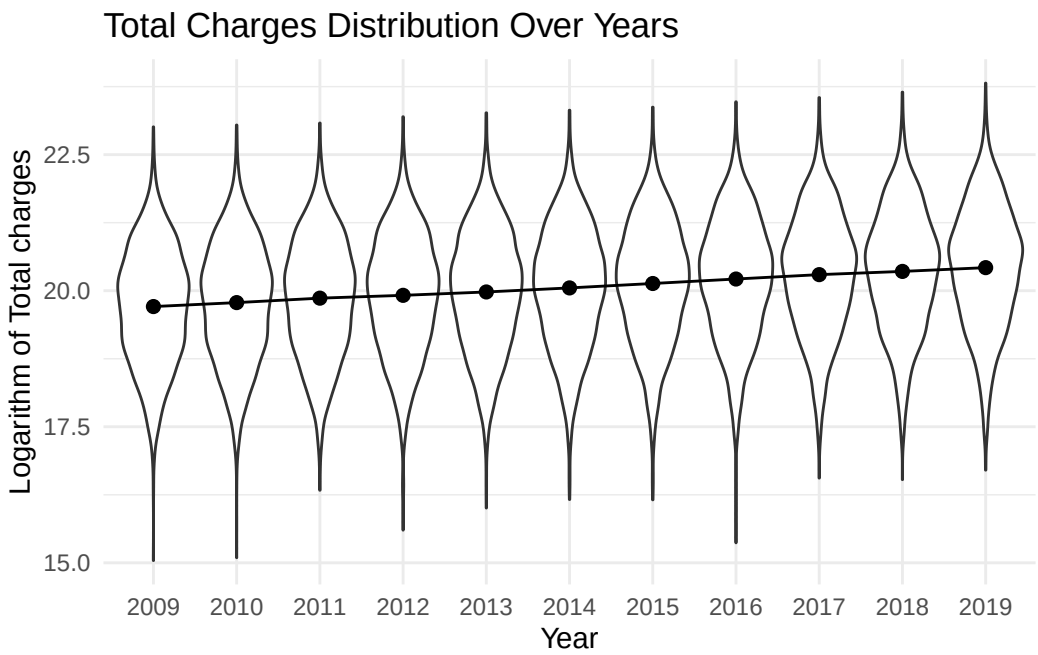
After removing/combining multiple reports, the number of unique hospital IDs (Medicare provider numbers) is about 6000 in the data.

Table 1: Number of Unique Hospitals per Year (2009–2019)

Year	Unique Hospitals
2009	6100
2010	6103
2011	6097
2012	6140
2013	6066
2014	6064
2015	6051
2016	6091
2017	6083
2018	6042
2019	6035

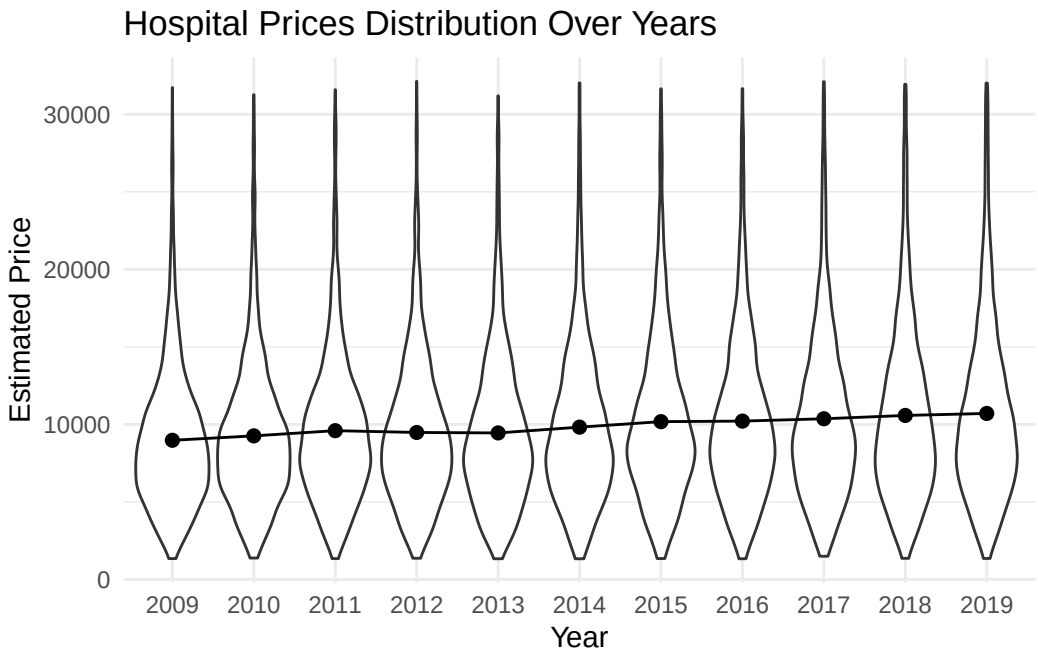
Problem 3

From 2009-2019, the distribution of total charges (tot_charges in the data) in each year is increasing, as can be seen. Note, the log y-axis scale has been implemented to capture standardized, prominent effects.



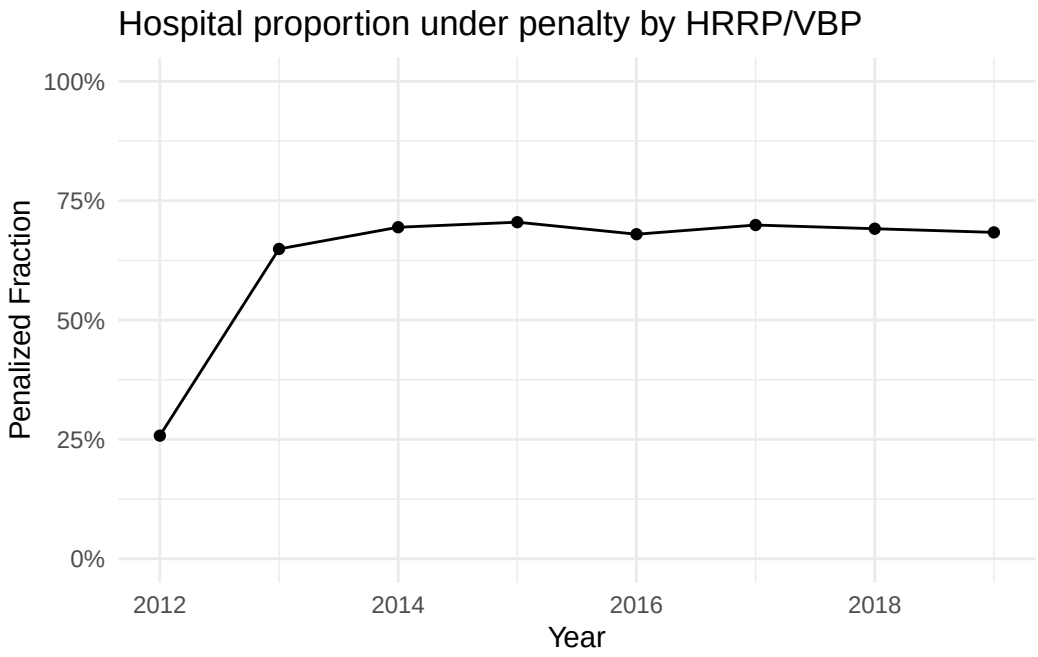
Problem 4

From 2009-2019, the distribution of estimated prices in each year has again been mildly increasing, approaching the $10,000 mark.



Problem 5

The proportion of hospitals under penalty by HRRP/VBP stays relatively at the same level of apprximately 70% throughout the majority of years.



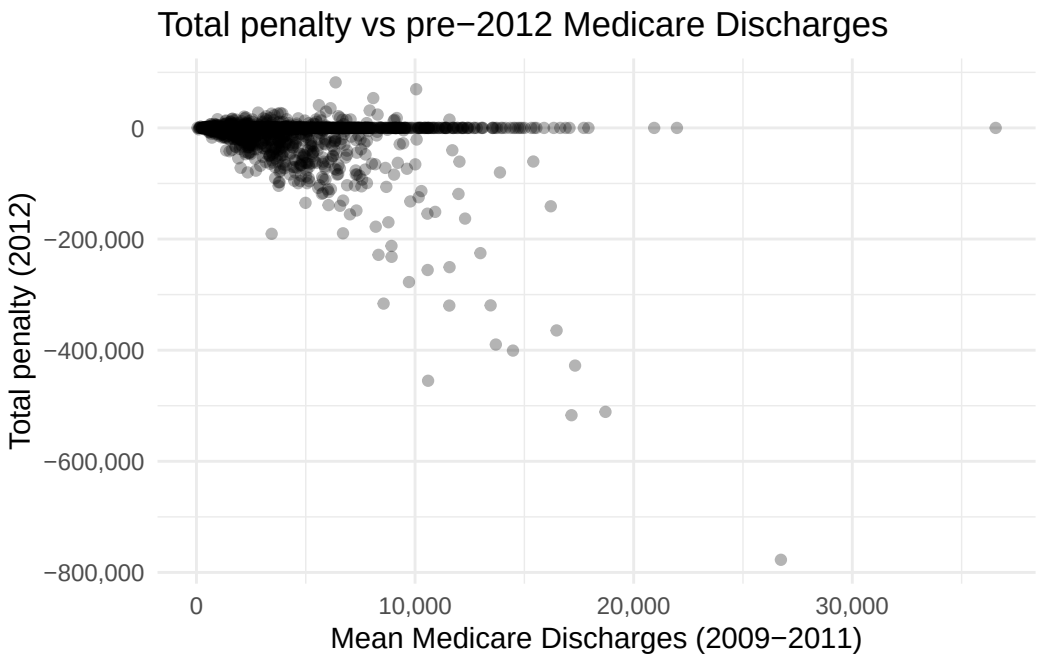
Problem 6

Table 2: OLS Estimates of Price Change on Penalty Status

Statistic	OLS 1	OLS 2	OLS 3
Coefficients			
Intercept	219.448	201.423	40.825
Net Penalty	-0.004	-0.004	-0.003
Avg Beds	0	0	0
Avg Medicaid	0	0	0
Model Fit			
Num.Obs.	2393.000	2393.000	2393.000
R2	0.003	0.005	0.012

We need the estimated effect on the net penalty is not significant, as observed from the coefficients. Further, average beds also maintain a low, yet similar estimated effect size of around 0.06 or 0.07. Finally, when average Medicaid discharges are included, we see the most prominent significance with the lowest p-value and largest estimate magnitude.

Problem 7



The negative relationship between net penalty and pre-2012 Medicare discharges observed here signifies that the hospitals with a larger percentage of Medicare patients happen to receive larger penalties.

Problem 8

Table 3: First Stage and Reduced Form Estimates

	First Stage			Reduced Form		
Statistic	**FS 1**	**FS 2**	**FS 3**	**RF 1**	**RF 2**	**RF 3**
Coefficients						
Intercept	6132.041	6129.816	5568.752	172.394	167.446	114.175
Avg Medicare	0	0	0	0	0	0
Avg Beds	0	0	0	0	0	0
Avg Medicaid	0	0	0	0	0	0
Model Fit						
Num.Obs.	2393.000	2393.000	2393.000	2393.000	2393.000	2393.000
R2	0.115	0.115	0.119	0.001	0.003	0.010

We note that in the first stage, net penalty has the largest effect and most prominent statistical significance with average Medicare discharges, showing Medicare discharge as a good instrumental variable. Whereas in reduced form, there still exists a nonzero relationship with average Medicare discharges, thus, it holds though not vividly; rather, the specification with average beds recovers the most statistically significant p-value effect.

Problem 9

Table 4: IV Estimates of Price Change on Net Penalty (Instrument: Avg Medicare Discharges)

Statistic	IV 1	IV 2	IV 3
Coefficients			
Intercept	204.384	194.003	87.131
Net Penalty (IV)	-0.005	-0.004	0.005
Avg Beds	0	0	0
Avg Medicaid	0	0	0
Model Fit			
Num.Obs.	2393.000	2393.000	2393.000
R2	0.002	0.005	-0.002

Here as well, we note that with no significant result in the first specification, the second specification with average beds becomes statistically significant, and maintains the same effect size in the third specification when included alongside average Medicaid discharges; however, yet again, average Medicaid discharges give the most prominent statistical significance with the largest effect size magnitude.

Problem 10

LATE accounts for changes in the neighborhood of a policy decision and not throughout the context window. If a policy is implemented starting Year X, it would be misleading to include statistics of years prior to that, when such numbers would be unrelated. IV acts similarly to regression discontinuity in the sense that both give an essence of LATE. Thus, here, the overall ATE would be polluted by extrapolating too much beyond the policy or decision implementation period, thereby giving noisy or biased results. Here, IV finally captures LATE estimates of net penalties for price changes in hospitals that had penalized status from larger Medicare patient proportions. Thus, it would not be completely rational to extrapolate it to penalized hospitals that weren't subject to large Medicare patient intake.